Solution 1

Problem 1 — Let X be a martingale (resp. supermartingale) on some filtered probability space, and let T be an a.s. finte stopping time. Prove that $\mathbb{E}[X_T] = \mathbb{E}[X_0]$ (resp. $\mathbb{E}[X_T] \leq \mathbb{E}[X_0]$) if either one of the following conditions holds:

1. The process X is bounded ($\exists M > 0 : \forall n \geq 0, |X_n| \leq M$ a.s.).
2. The process X has bounded increments ($\exists M > 0 : \forall n \geq 0, |X_{n+1} - X_n| \leq M$ a.s.) and $\mathbb{E}[T] < \infty$.

The supermartingale case immediately implies the martingale case, so we only prove the supermartingale case. First off, if T were bounded, then the Bounded Optional Stopping Theorem would conclude. In particular, $\mathbb{E}[X_{T\wedge n}] \leq \mathbb{E}[X_0]$ for any constant $n \geq 0$. So all we want to show is that $\mathbb{E}[X_{T\wedge n}] \to \mathbb{E}[X_T]$. All we have to do is rip through **Fatou-Lebesgue** by providing an integrable function which dominates all $X_{T\wedge n}$ almost surely.

In the first case, $X_{T\wedge n}$ is dominated by M almost surely, so we conclude.

In the second case, we need to use the condition $\mathbb{E}[T] < \infty$ by expanding in terms of increments. We know

$$X_{T\wedge n} = X_0 + \sum_{t=0}^{n-1} (X_{t+1} - X_t) \mathbb{1}_{t \leq T}$$

thus

$$\begin{aligned} |X_{T\wedge n}| &\leq |X_0| + \sum_{t\geq 0} |X_{t+1} - X_t| \, \mathbb{1}_{t\leq T} \\ &\leq |X_0| + M \sum_{t \geq 0} \mathbb{1}_{t \leq T} \\ &\leq |X_0| + M\mathbb{E}[T] < \infty \end{aligned}$$

The RHS of the top line is independent of n, and is our dominating function. We conclude.

Solution 2

Problem 2 — Let T be an $(\mathscr{F}_n, n \geq 0)$-stopping time such that for some integer $N > 0$ and $\epsilon > 0$, $\mathbb{P}[T \leq N + n | \mathscr{F}_n] \geq \epsilon$, for every $n \geq 0$. Show that $\mathbb{E}[T] < \infty$.

By definition, the condition says that for all measurable $A \in \mathscr{F}_n$,

$$\mathbb{P}[A \cap \{T \leq N + n\}] \geq \epsilon \mathbb{P}[A].$$

Sure, let's pick $A = \{T > n\}$! Then,

$$\mathbb{P}[n < T \leq N + n] \geq \epsilon \mathbb{P}[T > n].$$

Then,

$$\begin{aligned}
\mathbb{E}[T] &= \sum_{n \geq 0} \mathbb{P}[T > n] \\
&\leq \frac{1}{\varepsilon} \sum_{n \geq 0} \mathbb{P}[n < T \leq N + n] \\
&\leq \frac{1}{\varepsilon} \sum_{0 \leq i < N} \sum_{n \geq 0} \mathbb{P}[n \cdot N + i < T \leq (n+1) \cdot N + i] \\
&\leq \frac{1}{\varepsilon} \sum_{0 \leq i < N} \mathbb{P}[i < T] \\
&\leq \frac{1}{\varepsilon} \sum_{0 \leq i < N} 1 < \infty
\end{aligned}$$

Solution 3

Problem 3 — Let $(X_n, n \geq 0)$ be a sequence of $[0,1]$-valued random variables, which satisfy the following property. First, $X_0 = a$ a.s. for some $a \in (0,1)$ and for $n \geq 0$,

$$P\left\{X_{n+1} = \frac{X_n}{2}\middle|\mathscr{F}_n\right\} = 1 - X_n = 1 - P\left\{X_{n+1} = \frac{X_n+1}{2}\middle|\mathscr{F}_n\right\},$$

where $\mathscr{F}_n = \sigma(X_k, 0 \leq k \leq n)$. Here, we have denoted $\mathbb{P}[A|G] = \mathbb{E}[1_A|G]$.

1. Prove that $(X_n, n \geq 0)$ is a martingale that converges in L^p for every $p > 1$.
2. Check that $\mathbb{E}[(X_{n+1} - X_n)^2] = \mathbb{E}[X_n(1 - X_n)]/4$. Then determine $\mathbb{E}[X_\infty(1 - X_\infty)]$ and deduce the law of X_∞.

For the first part, by Doob's Martingale Convergence Theorem, it suffices to show that X_n is bounded in L^p for every $p > 1$. But this is clear, since $X_n \in [0,1]$, so $\mathbb{E}[|X_n|^p] \leq 1$ uniformly in n. As a bonus, we know that this limit X_∞ is also an almost-sure limit, and $X_n = \mathbb{E}[X_\infty|\mathscr{F}_n]$ almost surely for all $n \geq 0$.

For the second part, we compute

$$\begin{aligned}
\mathbb{E}[(X_{n+1} - X_n)^2] &= \mathbb{E}[\mathbb{E}[(X_{n+1} - X_n)^2|\mathscr{F}_n]] \\
&= \mathbb{E}\left[X_n\left(\frac{X_n+1}{2} - X_n\right)^2 + (1 - X_n)\left(\frac{X_n}{2} - X_n\right)^2\right] \\
&= \frac{1}{4}\mathbb{E}[X_n(1 - X_n)^2 + (1 - X_n)X_n^2] \\
&= \frac{1}{4}\mathbb{E}[X_n(1 - X_n)].
\end{aligned}$$

Now, we know that $X_n \to X_\infty$ almost surely, so by the Dominated Convergence Theorem,

$$\mathbb{E}[X_\infty(1 - X_\infty)] = \lim_{n\to\infty}\mathbb{E}[X_n(1 - X_n)] = 4\lim_{n\to\infty}\mathbb{E}[(X_{n+1} - X_n)^2] = 0$$

where the last equality is because X_n converge in L^2.

However, $\mathbb{E}[X_\infty(1 - X_\infty)]$ is nonnegative on $[0,1]$, with equality at $\{0,1\}$ only. Thus, X_∞ is almost surely 0 or 1. X is also UI, thus by optional stopping theorem, $\mathbb{E}[X_\infty] = \mathbb{E}[X_0] = a$. Thus, $\mathbb{P}(X_\infty = 1) = a$ and $\mathbb{P}(X_\infty = 0) = 1 - a$.

Solution 4

Let $(X_n, n \geq 0)$ be a sequence of independent and identically distributed real-valued integrable random variables. We let $S_n = X_1 + \cdots + X_n$ (with $S_0 = 0$) be the associated random walk and T an $(\mathscr{F}_n)$-stopping time, where $\mathscr{F}_n = \sigma(X_k, k \leq n)$.

Problem 4.1 — Show that if the random variables X_i are non-negative, then $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X_1]$.

Idea (Main idea) — This problem is very interesting! The key idea is that $\mathbb{E}[X_i \mathbb{1}_{T=n}]$ does *not* have to equal $\mathbb{E}[X_i]\mathbb{P}(T = n)$, because T is not necessarily independent of X_i. However, we notice that postponing T does not change the fact that we wil *eventually* count X_i; symbolically, we really care about $\mathbb{E}[X_i \mathbb{1}_{T \geq i}]$. Flipping this to $\mathbb{E}[X_i \mathbb{1}_{T<i}]$, we observe that $\{T < i\}$ and X_i really are independent, so we can compute everything.

If $\mathbb{E}[X_1] = 0$, then $X_1 = 0$ almost surely and the problem is a bit stupid; assume otherwise. Now, by monotone convergence or whatever,

$$\mathbb{E}[S_T] = \sum_{n=0}^{\infty} \mathbb{E}[S_n \mathbb{1}_{T=n}] + \mathbb{E}[S_\infty \mathbb{1}_{T=\infty}].$$

If $\mathbb{P}(T = \infty) > 0$, then $\mathbb{E}[X_1]\mathbb{E}[T] = \infty$, and the LHS is also ∞ (because S_∞ is also infinite almost surely, if you interpret this to mean the almost-sure limit; this guy might also be undefined so whatever). So we can assume that $\mathbb{P}(T = \infty) = 0$. Then, we observe the LHS is just

$$\sum_{n=0}^{\infty} \sum_{i \leq n} \mathbb{E}[X_i \mathbb{1}_{T=n}]$$

By Fubini's theorem we can interchange the sums.

$$\sum_{i=0}^{\infty} \sum_{n \geq i} \mathbb{E}[X_i \mathbb{1}_{T=n}]$$

Now, we observe that $\mathbb{P}(T \geq i) = 1 - \mathbb{P}(T \leq i - 1)$ which lives in $\mathscr{F}_{i-1}$ and is thus independent of X_i. Thus, we can write

$$= \sum_{i=0}^{\infty} \mathbb{E}[X_i]\mathbb{P}(T \geq i) = \mathbb{E}[X_1]\mathbb{E}[T]$$

as desired.

Problem 4.2 — Show that if $\mathbb{E}[T] < \infty$, then $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X_1]$.

First off the fact that $\mathbb{E}[T]$ exists means that $\mathbb{P}(T = \infty) = 0$.

Now, we repeat the above proof with $[0, \infty]$ replaced by $\mathbb{R}$. Consider the partial sums

$$\sum_{n=0}^{m} S_n \mathbb{1}_{T=n} \xrightarrow{m \to \infty} S_T$$

which converge almost surely. We need to show the expectation of the absolute value of the LHS is bounded so we can apply Dominated Convergence.

Of course, this is just the previous problem, because

$$\mathbb{E}\left[\left|\sum_{n=0}^{m} S_n \mathbb{1}_{T=n}\right|\right] \leq \sum_{n=0}^{\infty} \mathbb{E}[\tilde{S}_n \mathbb{1}_{T=n}] \leq \mathbb{E}[T]\mathbb{E}[|X_1|] < \infty$$

where $\tilde{S} = \sum_{i=0}^{n} |X_i|$; note that $|X_i|$ is integrable if X_i is.

Great! So we really care about the sum

$$\begin{aligned}
\sum_{n=0}^{\infty} \mathbb{E}[S_n \mathbb{1}_{T=n}] &= \sum_{n=0}^{\infty} \sum_{i \leq n} \mathbb{E}[X_i \mathbb{1}_{T=n}] \\
&= \sum_{i=0}^{\infty} \sum_{n \geq i} \mathbb{E}[X_i \mathbb{1}_{T=n}] \\
&= \sum_{i=0}^{\infty} \mathbb{E}[X_i]\mathbb{P}(T \geq i) \\
&= \mathbb{E}[X_1]\mathbb{E}[T]
\end{aligned}$$

by the same arguments as the first time; use Tonelli rather than Fubini.

Problem 4.3 — Suppose that $\mathbb{E}[X_1] = 0$ and set $T_a = \inf\{n \geq 0 : S_n \geq a\}$, for some $a > 0$. Show that $\mathbb{E}[T_a] = \infty$.

Applying the previous problem to $T = T_a$, we observe that if $\mathbb{E}[T_a] < \infty$ (such that $T_a < \infty$ almost surely) then $\mathbb{E}[S_T] = \mathbb{E}[T_a]\mathbb{E}[X_1] = 0$. However, in this case $S_T \geq a$ almost surely thus $\mathbb{E}[S_T] \geq a$, contradiction.

Thus $\mathbb{E}[T_a] = \infty$.

Problem 4.4 — Suppose that $\mathbb{P}[X_1 = +1] = 2/3 = 1 - \mathbb{P}[X_1 = -1]$ and set $T_a = \inf\{n \geq 0 : S_n \geq a\}$, for some $a > 0$. Find $\mathbb{E}[T_a]$. (You cannot assume that $\mathbb{E}[T_a] < \infty$.).

Well, if we knew that $\mathbb{E}[T_a] < \infty$, then $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X_1] = \frac{1}{3}\mathbb{E}[T]$; of course, in this case, $\mathbb{E}[S_T] = \lceil a \rceil$, so we would find $\mathbb{E}[T] = 3\lceil a \rceil$.

Okay, let's just compute an upper bound on $\mathbb{E}[T_a]$! The probability that T_a occurs after timestep $n \gg a$ is the probability that at least $\frac{1}{2}n - \frac{1}{2}a$ of the X_i are -1. This binomial sum is

$$\sum_{k=\frac{1}{2}n-\frac{1}{2}a}^{n} \binom{n}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{n-k}$$

Now, the claim is that individual terms $\binom{n}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{n-k}$ are decreasing; this is true for sufficiently large n, because the ratio of successive $\binom{n}{k}$ increases by at most $1 + O(\frac{a}{n})$ while the ratio of successive $\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{n-k}$ is exactly $1/2$.

Thus, this entire sum is bounded above by

$$\left(\frac{1}{2}n + \frac{1}{2}a\right)\binom{n}{\frac{1}{2}n - \frac{1}{2}a}\left(\frac{1}{3}\right)^{\frac{1}{2}n-\frac{1}{2}a}\left(\frac{2}{3}\right)^{\frac{1}{2}n+\frac{1}{2}a} \leq 2^n \cdot \left(\frac{2}{9}\right)^{n/2} = \left(\frac{2\sqrt{2}}{3}\right)^n$$

where I have dropped subleading terms in a.

Okay, fine! Now, recalling $\mathbb{E}[T] = \sum_{n=0}^{\infty} \mathbb{P}(T > n)$, we can truncate a finite prefix and bound the remaining terms by

$$\sum_{n=k}^{\infty} \left(\frac{2\sqrt{2}}{3} \right)^n < \infty$$

This concludes.

Alternatively, say something about a Chernoff bound to kill the problem.

Solution 5

Problem 5 — Suppose that $X_1, X_2, \cdots$ are independent random variables with $\mathbb{P}[X = +1] = p, \mathbb{P}[X = -1] = q$, where $p \in (0,1)$, $q = 1 - p$ and $p \neq q$. Suppose that a and b are integers with $0 < a < b$. Define $S_n = a + X_1 + \cdots + X_n$, $T = \inf\{n : S_n = 0 \text{ or } S_n = b\}$. Let $\mathscr{F}_n = \sigma(X_1, \cdots, X_n)$. Prove that

$$M_n = \left(\frac{q}{p}\right)^{S_n} \qquad \text{and} \qquad N_n = S_n - n(p-q)$$

define martingales M and N. Deduce the values of $\mathbb{P}[S_T = 0]$ and $\mathbb{E}[T]$.

First, we show M and N are martingales. Indeed,

$$\begin{aligned}
\mathbb{E}[M_{n+1}|\mathscr{F}_n] &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}} \middle| \mathscr{F}_n\right] \\
&= \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_n + X_{n+1}} \middle| \mathscr{F}_n\right] \\
&= \left(\frac{q}{p}\right)^{S_n} \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_{n+1}} \middle| \mathscr{F}_n\right] \\
&= \left(\frac{q}{p}\right)^{S_n} \left(\frac{p}{q} \cdot \mathbb{P}[X_{n+1} = -1] + \frac{q}{p} \cdot \mathbb{P}[X_{n+1} = +1]\right) \\
&= M_n \left(\frac{p}{q} \cdot q + \frac{q}{p} \cdot p\right) \\
&= M_n
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[N_{n+1}|\mathscr{F}_n] &= \mathbb{E}\left[S_{n+1} - (n+1)(p-q) \middle| \mathscr{F}_n\right] \\
&= \mathbb{E}\left[S_n + X_{n+1} - (n+1)(p-q) \middle| \mathscr{F}_n\right] \\
&= S_n - n(p-q) + \mathbb{E}[X_{n+1}] - (p-q) \\
&= N_n
\end{aligned}$$

For M, observe that $S_{T \wedge n}$ is a bounded martingale thus optional stopping theorem applies and $\mathbb{E}[M_T] = \mathbb{E}[M_0] = \left(\frac{q}{p}\right)^a$. For N, by arguments identical to 4d we know $\mathbb{E}[T] < \infty$, plus the increments are bounded thus optional stopping theorem applies and $\mathbb{E}[N_T] = \mathbb{E}[N_0] = a$.

On the other hand, we know that $S_T \in \{0, b\}$ almost surely. Thus we can extract the probabilities:

$$\mathbb{P}(S_T = 0) \cdot \left(\frac{q}{p}\right)^0 + \mathbb{P}(S_T = b) \cdot \left(\frac{q}{p}\right)^b = \left(\frac{q}{p}\right)^a$$

and

$$\mathbb{P}(S_T = 0) + \mathbb{P}(S_T = b) = 1$$

thus

$$\mathbb{P}(S_T = 0) + (1 - \mathbb{P}(S_T = 0)) \cdot \left(\frac{q}{p}\right)^b = \left(\frac{q}{p}\right)^a$$

and

$$\mathbb{P}(S_T = 0) = \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^b}{1 - \left(\frac{q}{p}\right)^b}.$$

We also know

$$\mathbb{P}(S_T = 0) \cdot 0 + \mathbb{P}(S_T = b) \cdot b - \mathbb{E}[T](p - q) = a$$

thus

$$\mathbb{E}[T] = \frac{\mathbb{P}(S_T = b) \cdot b + a}{p - q}$$

Solution 6

Problem 6 — Let $f : [0,1] \to \mathbb{R}$ be Lipschitz, that is, suppose that, for some $K < \infty$ and all $x, y \in [0,1]$

$$|f(x) - f(y)| \leq K|x - y|.$$

Denote by f_n the simplest piecewise linear function agreeing with f on $\{k2^{-n} : k = 0, 1, \cdots, 2^n\}$. Set $M_n = f_n'$. Show that M_n converges a.s. and in L^1 and deduce that f is the indefinite integral of a bounded function.

To be explicit, on the interval $[k \cdot 2^{-n}, (k+1) \cdot 2^{-n}]$, f_n is linear and agrees with f at the endpoints.

We claim that there is a Martingale hidden in here. Indeed, let $\Omega = [0,1]$; let our filtration be

$$\mathfrak{F}_n = \sigma(\{[k2^{-n}, (k+1)2^{-n}] : k = 0, 1, \cdots, 2^{n-1}\})$$

Then, M_n is $\mathfrak{F}_n$-measurable, and

$$\mathbb{E}[M_{n+1}|\mathfrak{F}_n] = \mathbb{E}[f_{n+1}'|\mathfrak{F}_n] = f_n'$$

(this is true because it is true on each segment $[k2^{-n}, (k+1)2^{-n}]$).

Thus M_n is a martingale! But it is also bounded by $\pm K$, because f is Lipschitz. Thus M_n is L^1 bounded and uniformly integrable. Thus, by Doob's Martingale Convergence Theorem, M_n converges almost surely and in L^1 to a limit M_∞, such that $M_n = \mathbb{E}[M_\infty|\mathfrak{F}_n]$ almost surely. Clearly $|M_\infty| \leq K$ almost surely.

Now, the claim is that the indefinite integral of M_∞ is f, i.e.

$$f(y) - f(0) = \mu\left(M_\infty \mathbb{1}_{x \leq y}\right)$$

almost everywhere. First let me show this for all dyadic $y = k \cdot 2^{-n}$. Indeed,

$$\begin{aligned}
\mu\left(M_\infty \mathbb{1}_{x \leq k \cdot 2^{-n}}\right) &= \sum_{i=0}^{k-1} \mu\left(M_\infty \mathbb{1}_{i \cdot 2^{-n} \leq x < (i+1) \cdot 2^{-n}}\right) \\
&= \sum_{i=0}^{k-1} \mathbb{E}\left[\mathbb{E}[M_\infty|\mathfrak{F}_n] \mathbb{1}_{i \cdot 2^{-n} \leq x < (i+1) \cdot 2^{-n}}\right] \\
&= \sum_{i=0}^{k-1} \mathbb{E}\left[M_n \mathbb{1}_{i \cdot 2^{-n} \leq x < (i+1) \cdot 2^{-n}}\right] \\
&= \sum_{i=0}^{k-1} f((i+1) \cdot 2^{-n}) - f(i \cdot 2^{-n}) \\
&= f(k \cdot 2^{-n}) - f(0)
\end{aligned}$$

For general y, we can take a dyadic sequence y_n approaching y from the left. Then, $\mu(M_\infty \mathbb{1}_{x \leq y}) = \lim \mu(M_\infty \mathbb{1}_{x \leq y_n})$ by dominated convergence (note that M_∞ is bounded), $\mu(M_\infty \mathbb{1}_{x \leq y_n}) = f(y_n)$ for all y_n by our construction, and $\lim f(y_n) = f(y)$ by continuity.